



Problem-Solution Fit

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Farmers (small & medium scale) • Agricultural Officers • Agribusiness Companies	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited budget • Poor internet connectivity • Basic smartphones	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Traditional farming advice • Agricultural experts • Weather forecast apps Pros: Experienced guidance, easily available. Cons: Not personalized, time-consuming, less accurate.
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2. PROBLEMS / PAINS

+ ITS FREQUENCY [PR]

- Incorrect crop selection – High
- Unpredictable weather – High
- Low crop yield – High
- Lack of expert guidance – Medium

9. PROBLEM ROOT /

CAUSE [RC]

- Lack of accurate data
- Climate variability
- Poor soil analysis
- Limited access to modern farming technology

7. BEHAVIOR

+ ITS INTENSITY [BE]

- Frequently checks weather updates
- Consults local farmers
- Uses mobile apps occasionally
- High dependence on seasonal information

3. TRIGGERS TO ACT

[TR]

- Low profits
- Crop failure
- Weather changes
- Need for better crop planning

10. YOUR SOLUTION

[SL]

OptiCrop is an AI-powered crop recommendation system that analyzes soil, weather, and environmental conditions to recommend the most suitable crops, helping farmers improve productivity, reduce risks, and increase profits.

8. CHANNELS OF

BEHAVIOR [CH]

ONLINE :

- Mobile App
- Website
- WhatsApp
- Social Media

4. EMOTIONS

BEFORE / AFTER [EM]

Before: Confused, worried, stressed, uncertain.

After: Confident, satisfied, hopeful, informed.

OFFLINE:

- Agriculture Offices
- Farmer Meetings
- Local Markets
- Training Programs